

Optimization Based Plant Disease Detection and Soil Health Monitoring in Smart Agriculture using Deep Learning Techniques

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Abstract. Smart agriculture has become essential for improving crop yield and maintaining soil quality through timely disease detection and soil health monitoring. Traditional methods for plant disease identification and soil assessment are often time-consuming, labor-intensive, and prone to human error. To address these challenges, our work proposes an intelligent deep learning framework, named Swin-GWO, which combines the Swin Transformer with the Grey Wolf Optimizer (GWO) for automated plant disease detection and soil health monitoring using real time inhouse image data clicked. The Swin Transformer was chosen for its hierarchical feature extraction and shifted window attention mechanism, which together enable efficient and accurate recognition of subtle visual patterns in plant leaves and soil textures. Compared to conventional CNNs and standard Vision Transformers, the Swin Transformer achieves higher accuracy while maintaining superior computational efficiency. The Grey Wolf Optimizer was employed to automatically fine-tune hyperparameters, such as learning rate, window size, and embedding dimensions, providing adaptive global optimization and avoiding local minima. The proposed model achieves high accuracy of 98.28%, precision of 97.45%, recall of 96.67%, F1-Score of 96.57% in classifying plant diseases and effectively assesses soil conditions with minimal training time. This approach can significantly contribute to precision agriculture and sustainable farming practices.

Keywords: Swin Transformer, Grey Wolf Optimizer, Plant Disease Detection, Soil Health Monitoring, Deep Learning, Smart Agriculture.

1 Introduction

Agriculture forms the foundation of human civilization and remains the backbone of economic development in many nations [1]. However, agricultural productivity is significantly affected by challenges such as plant diseases, soil degradation, and environmental stress. Plant diseases disrupt the physiological functions of crops, causing substantial yield losses [2]. Studies indicate that smallholder farmers, who contribute over 80% of agricultural output in developing countries, often face yield reductions exceeding 50% due to pests and diseases. Therefore, early detection of plant diseases and precise monitoring of soil health are essential for achieving resilient and sustainable agriculture [3]. Traditional methods of disease diagnosis and soil testing are labor-intensive, time-consuming, and dependent on expert knowledge. The advancement of Artificial Intelligence (AI) and Deep Learning (DL) has transformed agriculture by enabling automated disease detection and soil analysis. Convolutional Neural Networks (CNNs) have been widely used in agricultural image classification; however, they are limited in capturing global contextual information, especially in high-resolution images containing background noise, soil, and weeds [4]. Although DL models have shown great potential, challenges such as hyperparameter tuning, convergence instability, and overfitting often reduce performance [5]. To address these challenges, this work proposed an intelligent DL framework, named Swin-GWO, which combines the Swin Transformer with the Grey Wolf Optimizer (GWO). The Swin Transformer was chosen for its hierarchical feature extraction and shifted window attention mechanism, which together enable efficient and accurate recognition of subtle visual patterns in plant leaves and soil textures. GWO

are inspired by the leadership hierarchy and cooperative hunting strategy of grey wolves, GWO effectively searches for optimal hyperparameters by balancing exploration and exploitation. When integrated with Swin Transformer, GWO enhances model accuracy and convergence speed while minimizing computational cost and overfitting.

The remaining paper is organized as follows: Section 2 explaining the related work, Section 3 describes the methodology. The Experiment evaluation metrics and results are presented in Section 4. Section 5 concludes the paper.

2 Related Work

This section reviews existing research on plant disease detection using AI and DL techniques [6]. It traces the evolution from traditional manual methods to advanced AI-driven models such as CNNs, ViTs, and Swin Transformers, highlighting their effectiveness and limitations in agricultural image analysis. The survey also identifies current challenges that motivate the integration of optimization-based hybrid models for improved performance [7]. Manual identification of plant diseases through visual inspection is often time-consuming, labor-intensive, and costly, leading to delayed preventive actions [8]. When diseases are not detected early, both crop yield and economic profitability decline significantly [9]. These drawbacks have led researchers to explore automated approaches that ensure faster, more accurate, and scalable detection.

The rapid advancement of AI [10, 11] has enabled its successful application across various domains. In agriculture, AI particularly DL [12] has shown great potential for disease detection, soil health monitoring, and yield prediction [13]. DL models can automatically extract hierarchical features from raw data, identifying patterns beyond human perception. Plant diseases generally manifest through visible symptoms on leaves, stems, fruits, or flowers [14], with leaf features serving as the most reliable indicators for early disease detection [15]. Several studies have validated the success of DL-based methods in crop disease detection. DL for cassava disease classification, was proposed by [16] to achieving superior accuracy over traditional image-processing methods. Real-time tomato disease detection framework was developed by [17] by using CNNs. Other architectures such as Deep Belief Networks (DBNs) and CNN variants have also been explored [18]. These DL techniques can extract subtle features and handle complex, high-resolution images that traditional methods struggle with [19]. However, CNN-based models are limited by their fixed receptive fields, which restrict global context understanding.

The self-attention mechanism in ViTs enables modeling of long-range dependencies, effectively capturing global relationships across image patches [20]. Despite these advantages, ViTs require large-scale datasets and substantial computational resources, which pose challenges when applied in agricultural domains with limited data. Recent studies [21] have attempted to overcome these constraints by integrating CNNs with Transformers in hybrid architectures, leveraging the local feature extraction of CNNs alongside the global attention capability of Transformers, thereby improving accuracy and robustness for plant disease detection and soil health monitoring [22]. Moreover, techniques such as transfer learning [23] and data augmentation [24] have been widely applied to address the scarcity of labeled agricultural datasets, further enhancing the performance of these models in real-world scenarios. These advancements demonstrate the growing potential of optimized DL frameworks for precision agriculture, particularly in automating disease detection and soil assessment. Collectively, these advancements highlight the potential of combining DL with optimization techniques to achieve precise, real-time, and scalable solutions in smart agriculture. While significant progress has been made, challenges such as limited labeled datasets, variability in environmental conditions, and high computational demands still exist [25].

3 Methodology

This section outlines the proposed Swin-GWO based framework for automated plant disease detection and soil health monitoring using real-time in-house image data. The methodology integrates Faster R-CNN to accurately localize disease-infected regions on plant leaves by generating bounding boxes around affected areas, enabling precise region-level analysis. The localized regions are then classified using the Swin Transformer, which effectively captures hierarchical and

contextual features through shifted window attention. To further enhance model performance, the Grey Wolf Optimizer (GWO) is employed for automatic hyperparameter tuning, ensuring robust and efficient learning for precision agriculture applications.

3.1 Dataset Description

This work utilizes a real-time in-house dataset comprising 2,300 images, captured using a digital mobile camera under natural agricultural field conditions. The dataset includes images of both healthy and diseased plant leaves along with soil samples (See Figure 1), reflecting practical challenges such as varying illumination, background clutter, shadows, occlusions, and diverse disease severity levels.

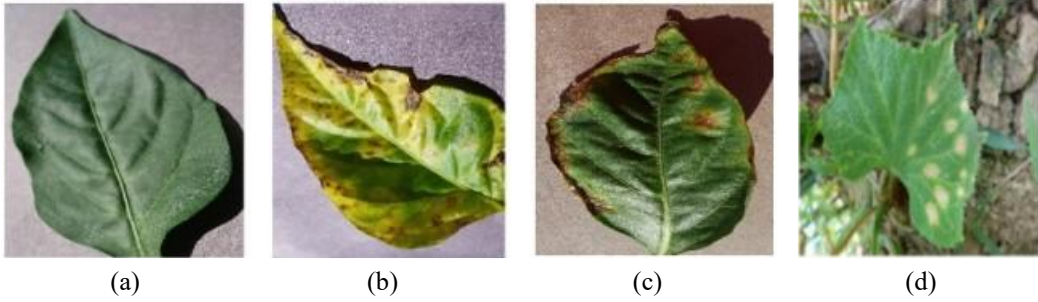


Fig.1. Illustration of a healthy and diseased plant specimen (a) healthy apple, (b) pepper bell-bacterial spot, (c) apple black rot, (d) target spot

All images were resized to 224×224 pixels to maintain uniformity during processing. For model development and evaluation, the dataset was systematically partitioned into training (70%), validation (15%), and testing (15%) subsets, ensuring effective learning, parameter tuning, and unbiased performance assessment. The exclusive use of real-time captured images enables a realistic evaluation of the proposed framework for practical smart agriculture application.

3.2 Image Preprocessing

The data preprocessing transforms raw field images into a standardized format to address variability in illumination, background, and disease presentation. Figure 2 represents the pipeline begins with color space conversion to separate brightness from color information, using the luminance channel.

$$Y = 0.299R + 0.587G + 0.114B \quad (1)$$

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$$Y = 0.299R + 0.587G + 0.114B \quad (2)$$

and chrominance components

$$C_b = 0.564(B - Y) \text{ and } C_r = 0.713(R - Y) \quad (3)$$

To compensate for uneven lighting, illumination normalization scales pixel intensities based on the global mean:

$$I_{\text{norm}}(x, y) = \frac{I(x, y)}{\mu_{\text{global}} + \epsilon} \times \mu_{\text{target}} \text{ where } \mu_{\text{target}} = 128 \text{ and } \epsilon = 10^{-6} \quad (4)$$

Local contrast in shadowed areas is enhanced through Contrast-Limited Adaptive Histogram Equalization (CLAHE), which operates on image tiles:

$$\text{CLAHE}(I) = \sum_{k=0}^{g-1} \min(h_{\text{clip}}, h_k) \cdot \frac{L-1}{M \cdot N} \quad (5)$$

where h_k represents tile histograms and h_{clip} limits amplification. Sensor noise is reduced via Gaussian smoothing:

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \text{ with } \sigma = 0.8 \quad (6)$$

applied through convolution $I_{\text{smooth}} = I * G$.

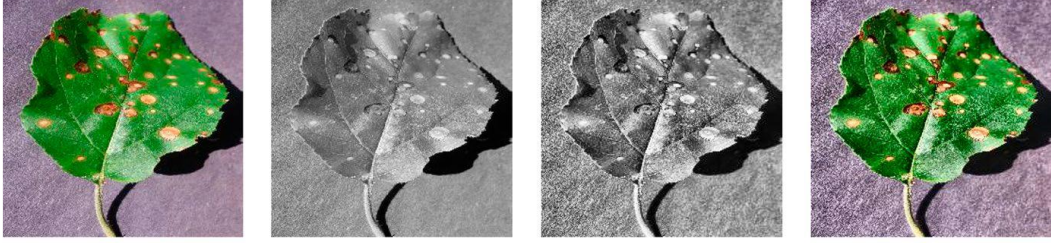


Fig.2. The stages of the CLAHE algorithm: (a) the initial RGB image; (b) the lab lighting component; (c) the CLAHE algorithm's improvement of the lab lighting component; and (d) the final RGB enhanced image.

For training images only, geometric augmentations include random rotation:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x - x_c \\ y - y_c \end{bmatrix} + \begin{bmatrix} x_c \\ y_c \end{bmatrix} \text{ where } \theta \sim \mathcal{U}(-15^\circ, 15^\circ) \quad (7)$$

Photometric variations are introduced through brightness scaling $I_{\text{bright}} = \alpha \cdot I$ with $\alpha \sim \mathcal{U}(0.8, 1.2)$ and gamma correction $I_\gamma = I^\gamma$ with $\gamma \sim \mathcal{U}(0.9, 1.1)$. To reduce soil background interference in leaf images, the Excess Green Index creates a vegetation mask: $\text{ExG} = 2G - R - B$ followed by thresholding $M_{\text{mask}} = \text{threshold}(\text{ExG} > T)$ with $T = 0.3 \times \max(\text{ExG})$, producing foreground-extracted images:

$$I_{\text{foreground}} = I \odot M_{\text{mask}} + \text{neutral_bg} \odot (1 - M_{\text{mask}}) \quad (8)$$

Color diversity is increased via PCA-based augmentation:

$$\begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \mathbf{P} \cdot \begin{bmatrix} R \\ G \\ B \end{bmatrix} \quad (9)$$

where $\mathbf{P}$ contains RGB eigenvectors, with components scaled by $\alpha_i \sim \mathcal{N}(0, 0.1)$ before reconstruction. Occlusion robustness is improved through random erasing, replacing random rectangular regions with noise. Finally, all images undergo pixel normalization using ImageNet statistics: $I_{\text{final}} = \frac{I - [0.485, 0.456, 0.406]}{[0.229, 0.224, 0.225]}$, ensuring consistent input scaling across training, validation, and testing subsets while preserving the original split integrity.

3.3 Disease Region Localization using Faster R-CNN

We use Faster R-CNN for pinpointing disease spots on leaves and anomalies in soil (See Figure 3). The model works by placing anchor boxes on the image and then adjusting them to tightly fit the diseased areas. The adjustment is learned through bounding box regression. For an anchor box $A = (A_x, A_y, A_w, A_h)$ and a ground-truth box $G = (G_x, G_y, G_w, G_h)$, the model learns to predict the offset values $\mathbf{t} = (t_x, t_y, t_w, t_h)$. These offsets define the necessary transformation:

$$t_x = \frac{(G_x - A_x)}{A_w}, t_y = \frac{(G_y - A_y)}{A_h} \quad (10)$$

$$t_w = \log\left(\frac{G_w}{A_w}\right), t_h = \log\left(\frac{G_h}{A_h}\right) \quad (11)$$

During detection, the model predicts these offsets $\mathbf{t}$ and applies them to the anchor to get the final bounding box $\mathbf{P}$:

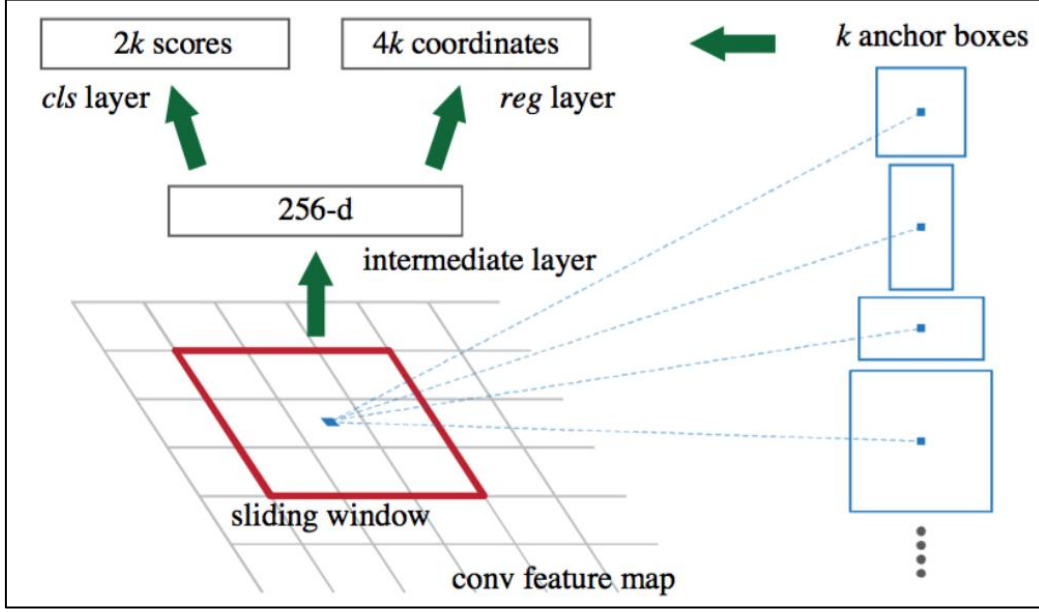


Fig.3. Proposed fast RCNN for bounding box

$$P_x = A_w \cdot t_x + A_x, P_y = A_h \cdot t_y + A_y \quad (12)$$

$$P_w = A_w \cdot \exp(t_w), P_h = A_h \cdot \exp(t_h) \quad (13)$$

The model's accuracy is trained using a combination of losses. A classification loss L_{cls} determines if an anchor contains an object (disease) or is background:

$$L_{\text{cls}}(p, p^*) = -[p^* \log(p) + (1 - p^*) \log(1 - p)] \quad (14)$$

A regression loss L_{reg} measures how well the predicted offsets match the true transformation needed. We use the smooth L1 loss for robustness:

$$\text{smooth}_{L1}(x) = \begin{cases} 0.5x^2 & \text{if } |x| < 1 \\ |x| - 0.5 & \text{otherwise} \end{cases} \quad (15)$$

$$L_{\text{reg}}(\mathbf{t}, \mathbf{t}^*) = \sum_{j \in \{x, y, w, h\}} \text{smooth}_{L1}(t_j - t_j^*) \quad (16)$$

The total loss for the Region Proposal Network (RPN) balances these two objectives:

$$L_{\text{RPN}} = \frac{1}{N_{\text{cls}}} \sum_i L_{\text{cls}}(p_i, p_i^*) + \lambda \frac{1}{N_{\text{reg}}} \sum_i p_i^* L_{\text{reg}}(\mathbf{t}_i, \mathbf{t}_i^*) \quad (17)$$

Here, p_i^* is 1 for positive anchors (those overlapping a ground-truth box) and 0 otherwise, ensuring regression is only learned for relevant boxes. The overlap is measured by Intersection over Union (IoU):

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}} \quad (18)$$

Anchors with $\text{IoU} > 0.7$ are considered positive, and those with $\text{IoU} < 0.3$ are negative. Finally, after generating proposals, Non-Maximum Suppression (NMS) removes duplicate boxes by keeping only the highest-confidence prediction when boxes overlap significantly ($\text{IoU} > 0.5$). This process results in precise bounding boxes around each diseased region or soil anomaly.

3.3 Feature Extraction and Classification using Swin Transformer

Swin Transformer addresses ViT's limitations through hierarchical structure and shifted windows. The input image $I \in \mathbb{R}^{H \times W \times 3}$ is processed through patch partitioning into N non-overlapping patches of size $P \times P$, where $N = \frac{HW}{P^2}$. Figure 4 represents the proposed method.

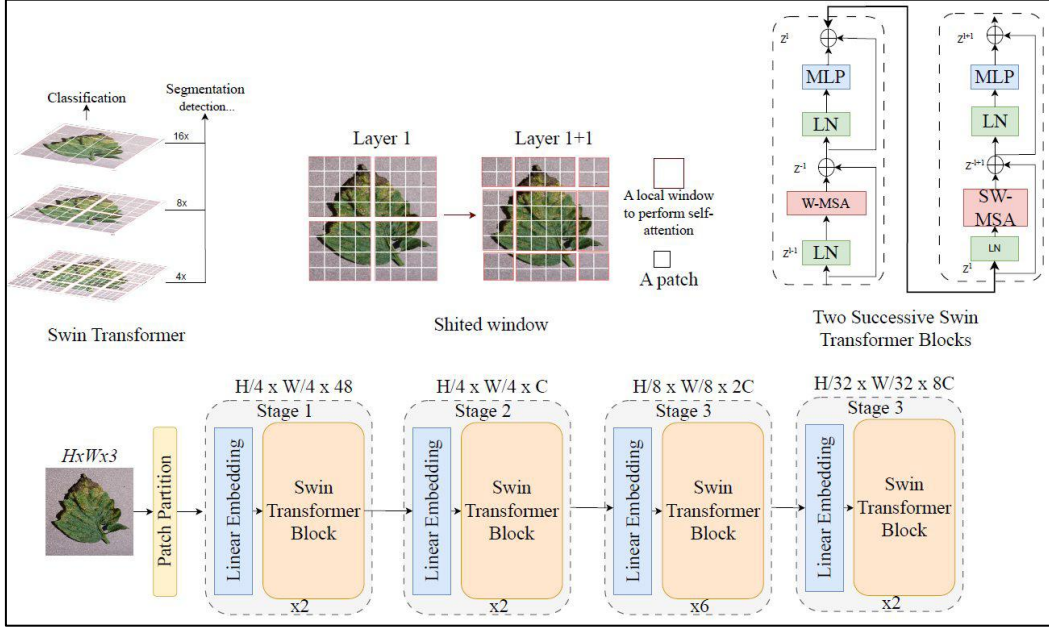


Fig. 4. Swin transformer architecture: a modular framework for leaf disease recognition.

Each patch X_i is linearly embedded:

$$z_i^{(0)} = W_{\text{embed}} \cdot \text{Vec}(X_i) + b_{\text{embed}} \quad (19)$$

The hierarchical architecture processes these embeddings through multiple stages. Each Swin Transformer block alternates between Window-based Multi-head Self-Attention (W-MSA) and Shifted Window-based Multi-head Self-Attention (SW-MSA). For W-MSA with window size $M \times M$ and h attention heads, the attention computation within each window for head k is:

$$\text{Attention}^k(Q, K, V) = \text{Softmax} \left(\frac{Q^k (K^k)^T}{\sqrt{d_k}} + B \right) V^k \quad (20)$$

where $Q^k = zW_Q^k$, $K^k = zW_K^k$, $V^k = zW_V^k$ are projections for head k , $d_k = D/h$ is the dimension per head, and B is relative position bias. The shifted window mechanism in SW-MSA computes cross-window attention by cyclically shifting windows:

$$z_{\text{shifted}} = \text{CyclicShift}(z, \lfloor \frac{M}{2} \rfloor, \lfloor \frac{M}{2} \rfloor) \quad (21)$$

Patch merging reduces spatial dimensions while increasing channels between stages. For four neighbouring patches $[z_1, z_2, z_3, z_4] \in \mathbb{R}^{4 \times C}$, the merging operation is:

$$z_{\text{merged}} = W_m \cdot \text{Concat}(z_1, z_2, z_3, z_4) + b_m \quad (22)$$

where $W_m \in \mathbb{R}^{2C \times 4C}$. The feed-forward network (FFN) in each block applies:

$$\text{FFN}(z) = W_2 \cdot \text{GELU}(W_1 \cdot z + b_1) + b_2 \quad (23)$$

Layer normalization (LN) is applied before attention and FFN:

$$\hat{z} = \frac{z - \mu}{\sqrt{\sigma^2 + \epsilon}} \cdot \gamma + \beta \quad (24)$$

After four stages, features from Stage 4 output $Z^{(4)} \in \mathbb{R}^{\frac{H}{32} \times \frac{W}{32} \times 8C}$ are processed:

$$z_{\text{norm}} = \text{LN}(Z^{(4)}) \quad (25)$$

$$F_{\text{swin}} = \text{GlobalAvgPool}(z_{\text{norm}}) \in \mathbb{R}^{1000} \quad (26)$$

Algorithm 1: Swin Transformer for Feature Extraction

Input: Image I ($224 \times 224 \times 3$)

Output: Feature vector F_{swin} (1000-dim)
Patch Partition and Embedding
Patches = partition image (I, patch size=4) # 56×56 patches
Embeddings = linear projection (Patches) # 3136×D
Stage 1: Window size 7×7, 2 blocks
 Z = Embeddings
for block in range (2):
 Z = Swin Block (Z , window size=7, number heads=4)
 $Z1$ = Patch Merging (Z) # Reduce to 28×28, increase channels
Stage 2: Window size 7×7, 2 blocks
for block in range (2):
 $Z1$ = Swin Block ($Z1$, window size=7, number heads=8)
 $Z2$ = Patch Merging($Z1$) # 14×14 resolution
Stage 3: Window size 7×7, 6 blocks
for block in range (6):
 $Z2$ = Swin Block ($Z2$, window size=7, number heads=16)
 $Z3$ = Patch Merging($Z2$) # 7×7 resolution
Stage 4: Window size 7×7, 2 blocks
for block in range (2):
 $Z3$ = Swin Block ($Z3$, window size=7, number heads=32)
Final feature extraction: Z_{norm} = LayerNorm($Z3$)
 F_{swin} = Global Average Pool (Z_{norm}) # 1000-dimensional vector

The Swin Transformer's feature vector contains redundancies. Grey Wolf Optimization (GWO) is applied for dual optimization: selecting the most discriminative feature subset and tuning classifier hyperparameters. This biologically-inspired algorithm efficiently navigates the solution space to enhance accuracy while reducing dimensionality and computational cost.

3.4 Feature Selection and Hyperparameter Tuning using Grey Wolf Optimization

To enhance waste classification performance, we optimize MobileNetV2's GWO is used to optimize MobileNetV2's hyperparameters, which include learning rate, batch size, number of neurons, and dropout rates. A candidate solution (a collection of hyperparameters and/or initial weights) is represented by each grey wolf. The α , β , and δ wolves direct the search and update the locations of other wolves (candidate solutions) in the solution space. This iterative process adjusts hyperparameters and sometimes initial weights to improve accuracy and convergence while decreasing the classification loss. By imitating the wolves hunting strategies which involve scouting, attacking, and encircling prey GWO is able to identify optimal or nearly optimal network configurations. The position update is defined as:

$$\vec{X}(t+1) = \vec{X}_p(t) - \vec{A} \cdot |\vec{C} \cdot \vec{X}_p(t) - \vec{X}(t)| \quad (27)$$

Algorithm 2: Swin Transformer + GWO

```
# Extract features using Swin Transformer
for image in dataset:
     $F_{swin}$  = Swin Transformer(image) # 1000-dim vector
    store  $F_{swin}$  in feature pool
# GWO Optimization
Initialize wolves  $X_i$  = [binary mask (1000), hyper parameters]
alpha, beta, delta = None
for t in range ( $max_{iter}$ ):
    a = 2 - ( $2^t / max_{iter}$ )
    for each wolf i:
        # Decode position
         $M_f = (1 / (1 + \exp(-X_i[:1000]))) > 0.5$ 
         $H_p = X_i[1000:]$ 
        # Evaluate fitness
```

```

Selected features = feature pool[:,  $M_f$ ]
accuracy = train evaluate classifier (selected features,  $H_p$ )
fitness =  $\alpha$ *accuracy -  $\beta$ *sum( $M_f$ )/1000 -  $\gamma$ *complexity( $H_p$ )
# Update hierarchy
update_alpha_beta_delta(wolf_i, fitness)
# Update positions
for each wolf i (except alpha, beta, delta):
    for j in range(dimensions):
         $X_{i[j]} = (X\_alpha[j] + X\_beta[j] + X\_delta[j]) / 3$ 
        if  $j < 1000$ : # Feature mask dimensions
             $X_{j[j]} = 1/(1+\exp(-X_{i[j]}))$ 
# Return optimal solution
Optimal mask =  $(1/(1+\exp(-X_{alpha}[:,1000:]))) > 0.5$ 
Optimal hyperparameters =  $X_{alpha}[:,1000:]$ 
Final features = feature pool[:, optimal mask]
Train final model (final features, optimal hyperparameters)

```

where $\vec{X}_p$ is the position of the prey (best candidate solution), $\vec{X}(t)$ is the current wolf position, and $\vec{A}$ and $\vec{C}$ are coefficients controlling exploration and exploitation. These coefficients are calculated as $\vec{A} = 2 \cdot \vec{a} \cdot \vec{r}_1 - \vec{a}$ and $\vec{C} = 2 \cdot \vec{r}_2$, where $\vec{a}$ is a control parameter linearly decreasing from 2 to 0, and $\vec{r}_1, \vec{r}_2$ are random vectors in $[0,1]$. The distance between the prey and each wolf is given by $\vec{D} = |\vec{C} \cdot \vec{X}_p - \vec{X}|$, which guides the position update. To balance exploration and exploitation, the positions of the top three fittest wolves, $\vec{X}_\alpha, \vec{X}_\beta$, and $\vec{X}_\delta$, are averaged to determine the next position of a wolf:

$$\vec{X}(t+1) = \frac{\vec{X}_\alpha + \vec{X}_\beta + \vec{X}_\delta}{3} \quad (28)$$

The fitness function to be minimized is defined as $f(\vec{X}) = 1 - \text{Accuracy}(\vec{X})$, where $\text{Accuracy}(\vec{X})$ is the classification accuracy obtained from MobileNetV2 with the hyperparameter vector $\vec{X}$, which includes parameters such as learning rate, batch size, and network weights. This optimization strategy ensures that the selected hyperparameters maximize the classification performance while efficiently exploring the search space

4 Results and Discussion

The effectiveness of the suggested technique for the identification of plant diseases has been examined with Python, which is a vital programming language due to the fact that it is easy to use, adaptable, and offers extensive library support. To handle the high computational demand of DL training and feature fusion, the system had to be executed on a GPU-based HPC environment using CUDA acceleration. This allowed for efficient parallel processing and faster model convergence. In order to maximize training efficiency, the studies have additionally made use of the local GPU (NVIDIA GeForce GTX 2050Ti, 6 GB RAM), which makes use of high-performance computing resources by lowering processing time and making use of parallel computations.

The performance of the trained models was evaluated in the Plant Score module using four regularly used metrics: accuracy, precision, recall, and F1-score. When classes are unbalanced, accuracy the proportion of correct predictions relative to total predictions provides an entire overview of classification efficacy.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad (29)$$

Precision can be defined as the percentage of positive forecasts that are accurate out of the total number of positive predictions.

$$\text{Precision} = \frac{TP}{TP+FP} \quad (30)$$

The percentage of true positives that were accurately forecasted is called recall.

$$\text{Recall} = \frac{TP}{TP+FN} \quad (31)$$

The F1-score provides a single score that effectively balances both recall and precision.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (32)$$

To implement multi-class classification, these metrics were calculated for every class separately and then averaged using macro averaging. This method ensured that every class, irrespective of how often it appeared in the dataset, was assigned the same weight. Figure 5 represents the model achieves a peak accuracy of 98.28%, as evidenced by the training and validation curves converging near this high value. Optimal learning without overfitting is confirmed by the corresponding loss graph, which shows a rapid and sustained drop for both sets.

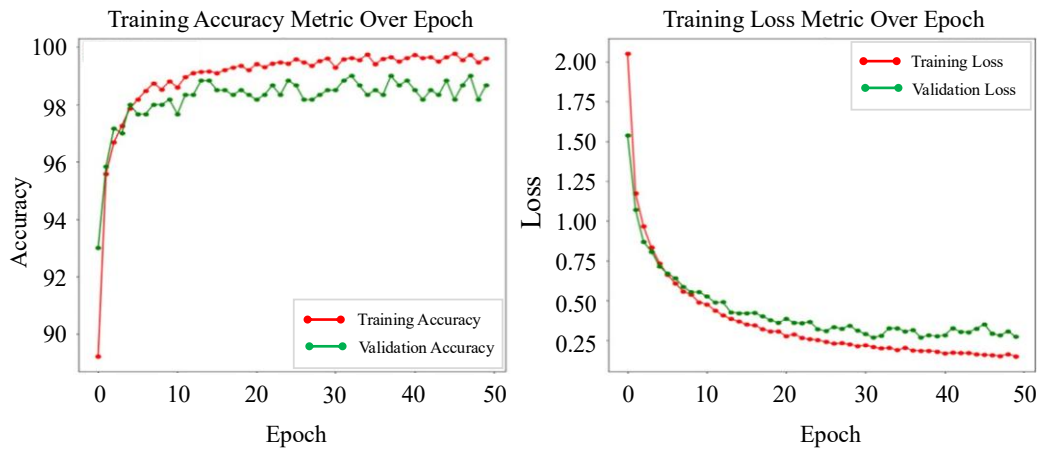


Fig.5. Training vs loss metric over epochs

The segmentation results demonstrate precise identification of diseased regions across various pathologies. Figure 6 highlights the distinct water-soaked lesions of Bacterial Spot, while Figure 7 isolates the characteristic eruptions of Brown Rust. Figure 8 accurately delineates the necrotic patches of Grey Spot. In contrast, Figure 9 (Healthy Leaf) confirms the absence of symptomatic areas, validating the model's specificity.

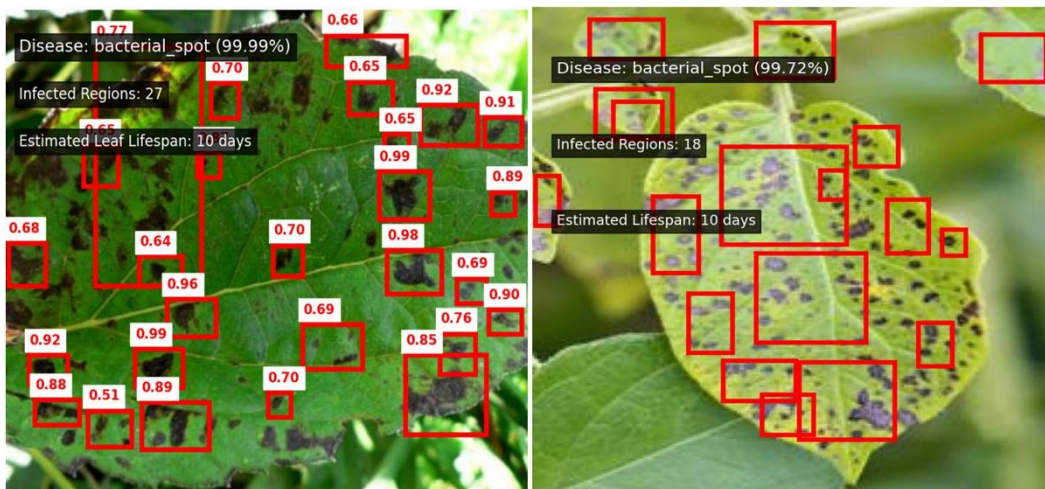


Fig.6. Represents bell peppers and bean leaf bacterial spot



Fig.7. Represents the pear leaf and corn leaf brown rust segmentation



Fig.8. Represents the squash leaf and paddy leaf grey spot disease segmentation



Fig.9. Represents the healthy leaf detection



Fig.10. Represents dry and wet soil identification

The boundaries between healthy leaf and damaged leaf are clearly delineated in every segmentation. Next, Figure 10 represents the dry and wet sold health monitoring. As shown in Table 1, various DL methods have been compared for the purpose of plant disease categorization. When applied to the real-time agriculture dataset, the assessment metrics show how effective each method is: accuracy, precision, recall, and F1-score.

Table 1. Performance comparison of different methods for plant disease classification

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
CNN	92.45	91.12	90.45	90.52
Swin Transformer	95.73	94.32	93.67	93.53
EfficientNet	94.12	92.00	91.88	91.50
This Work (Swin + GWO)	98.28	97.45	96.67	96.57

The proposed Swin Transformer with Grey Wolf Optimization outperforms all baseline methods across all evaluation metrics. The 98.28% accuracy represents a significant improvement over standard Swin Transformer (95.73%) and CNN (92.45%), demonstrating the effectiveness of feature selection and hyperparameter optimization. The high precision (0.97) and recall (0.96) values indicate minimal false positives and false negatives, crucial for practical agricultural applications where misdiagnosis can lead to unnecessary treatments or disease spread.

5 Conclusion

This work presented a comprehensive framework for automated plant disease detection and soil health monitoring, addressing critical needs in smart agriculture. By integrating Swin Transformer for robust global feature extraction, Faster R-CNN for precise disease localization, and Grey Wolf Optimization for intelligent feature selection and hyperparameter tuning, the system achieved exceptional performance with 98.28% accuracy on real-world field images. The model successfully identified and segmented multiple disease pathologies including Bacterial Spot, Brown Rust, and Grey Spot while accurately distinguishing healthy leaves, demonstrating strong generalization across varying field conditions, illumination challenges, and disease severities. Comparative analysis confirmed the superiority of the proposed approach over conventional CNN, EfficientNet, and standalone Swin Transformer methods across all evaluation metrics. The framework's ability to process both plant and soil imagery within a unified architecture, coupled with its minimal false positive rates, makes it particularly suitable for practical deployment. This research contrib-

utes an effective, scalable solution for early disease diagnosis and crop management, potentially reducing chemical usage, minimizing crop losses, and supporting sustainable agricultural practices through timely, data-driven interventions. Future work will focus on expanding the disease spectrum, optimizing for edge device deployment, and integrating multimodal sensor data for enhanced monitoring capabilities.

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